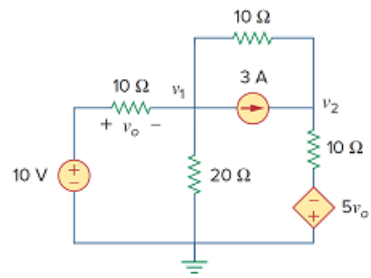


Problem

Determine v_1 and v_2 in the circuit of Fig. 3.71.

Figure 3.71:

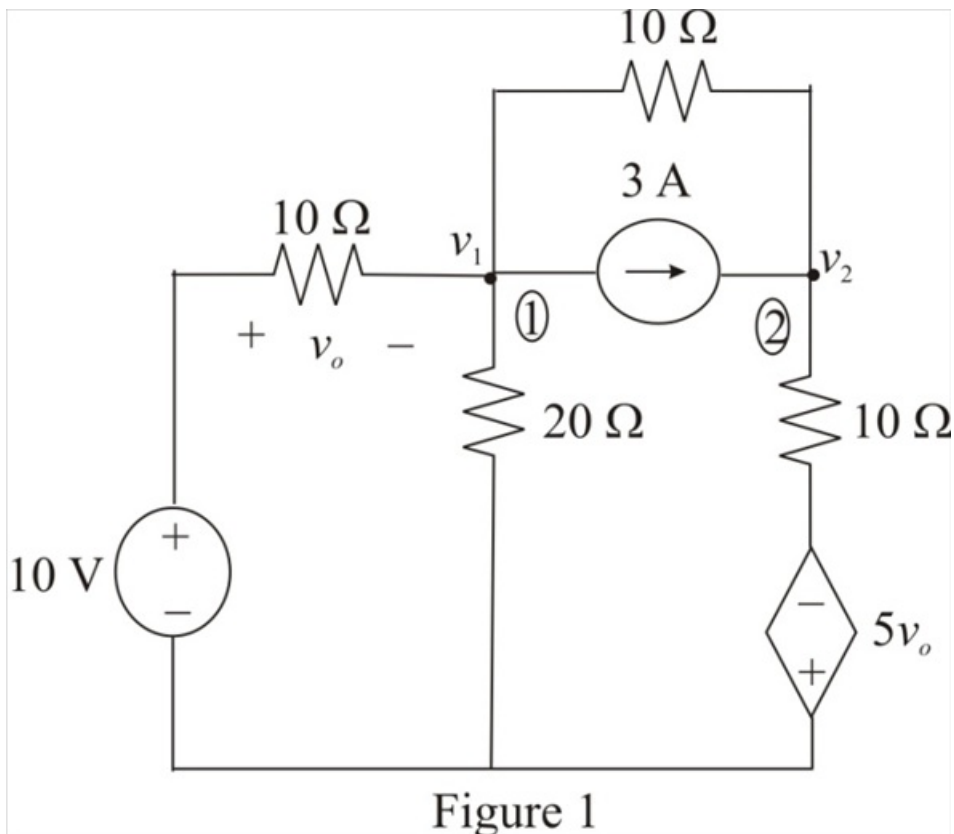


Step-by-step solution

Step 1 of 5

Refer to Figure 3.71 in the text book.

Identify the nodes and voltages in the circuit shown in Figure 1.



Step 2 of 5

Consider the expression for the voltage v_o .

$$v_o = 10 - v_1$$

Therefore, the expression for the voltage v_o is, $10 - v_1$.

Step 3 of 5

Apply nodal analysis at node 2.

$$\frac{v_2 - (-5v_o)}{10} - 3 + \frac{v_2 - v_1}{10} = 0$$

$$\frac{v_2 + 5v_o}{10} - 3 + \frac{v_2 - v_1}{10} = 0$$

$$\frac{v_2 + 5v_o - 30 + v_2 - v_1}{10} = 0$$

$$2v_2 + 5v_o - 30 - v_1 = 0$$

Substitute $10 - v_1$ for v_o .

$$2v_2 + 5(10 - v_1) - 30 - v_1 = 0$$

$$2v_2 + 50 - 5v_1 - 30 - v_1 = 0$$

$$2v_2 - 6v_1 + 20 = 0$$

$$v_2 - 3v_1 + 10 = 0$$

$$v_2 = 3v_1 - 10 \dots\dots (1)$$

Therefore, the expression for the voltage v_2 is, $3v_1 - 10$.

Step 4 of 5

Apply nodal analysis at node 1.

$$\frac{v_1 - 10}{10} + 3 + \frac{v_1 - v_2}{10} + \frac{v_1}{20} = 0$$

$$\frac{2v_1 - 20 + 60 + 2v_1 - 2v_2 + v_1}{20} = 0$$

$$2v_1 - 20 + 60 + 2v_1 - 2v_2 + v_1 = 0$$

$$5v_1 - 2v_2 - 20 + 60 = 0$$

$$5v_1 - 2v_2 + 40 = 0$$

Substitute $3v_1 - 10$ for v_2 .

$$5v_1 - 2(3v_1 - 10) + 40 = 0$$

$$-v_1 + 60 = 0$$

$$v_1 = 60 \text{ V}$$

Therefore, the value of the voltage v_1 is, 60 V.

Step 5 of 5

Recall equation (1).

$$v_2 = 3v_1 - 10$$

Substitute 60 V for v_1 .

$$v_2 = 3(60) - 10$$

$$= 180 - 10$$

$$= 170 \text{ V}$$

Therefore, the value of the voltage v_2 is, 170 V.